Sense HAT Image Control Application

ENGR 7 | Requirements Specification | October 25, 2026

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1 Problem statement

Develop a deployable MATLAB application that displays one image on the Raspberry Pi Sense HAT LED matrix. The user will rotate and vertically flip the image using the joystick, then press the center button to end the application. All images shown on the Sense HAT must be displayed using displayImage().

2 Equipment and input

Use the assigned classroom Raspberry Pi with its Sense HAT attached, a power supply, and MATLAB with the required Raspberry Pi support software. Joystick directions are defined with the Raspberry Pi HDMI port facing downward, as in the class template.

The supported input is an 8-bit RGB PNG. Choose a character or image with visible differences between its top, bottom, left, and right. The submitted image must be included with the source files.

3 Functional requirements

R1 Startup: During MATLAB development, accept the assigned Pi IP address and image file path as text. Connect to the Pi, create the Sense HAT object, and clear the LED matrix.

R2 Image display: Resize the complete image to 8 by 8 pixels using nearest-neighbor resizing. Keep its three RGB channels as an 8-by-8-by-3 uint8 array. Display the initial image and each transformed image using displayImage(mysh,hatimg).

R3 Joystick controls: Apply each operation to the current image, so rotations and flips can be combined.

Code	Joystick	Required behavior
0	Not pressed	Keep the current image unchanged.
1	Center	End the application and clear the LED matrix.
2	Left	Rotate 90 degrees counterclockwise.
3	Up	Flip vertically by reversing the image rows.
4	Right	Rotate 90 degrees clockwise.
5	Down	Flip vertically by reversing the image rows.

R4 Repetition: Holding a direction repeats its operation. Use a 0.1 s pause per loop; communication adds to this interval. Holding up or down therefore alternates between the two flipped states. Both directions perform the same vertical flip, following the class template.

4 Code organization and deployment

Use a main script for the application sequence, a while loop for repeated joystick input, a switch statement to select each operation, and short helper functions for setup, image loading, and orientation.

Source file	Responsibility
imageorientationmain.m	Read the inputs, call the helpers, display the image, repeat joystick actions, and clear the LEDs on exit.
senshat_setup.m	Create the Raspberry Pi and Sense HAT objects and clear the initial display.
loadAndConfigureImage.m	Read the PNG and resize it to the required RGB array.
orientImage.m	Return the rotated or flipped image for the selected joystick command.

R5 Deployment: Prepare the final application for the classroom Raspberry Pi deployment workflow. Include the selected image data in the deployed application or place the image on the target as required by that workflow. A Windows file path is only for desktop testing. Verify the target build and launch before presenting deployment as complete.

Document each file with its purpose, inputs, outputs, and short comments explaining the control loop and joystick mapping. For this version, use valid connection details and a supported RGB image; correct any MATLAB connection or file error before restarting the application.

5 Demonstration checklist

Use the same submitted image throughout. The expected outcomes below are acceptance checks, not a record of completed hardware tests.

1. Start the application and show the initial image on the Sense HAT. Explain the supported image format and resizing.

2. Leave the joystick untouched and confirm the image stays still. Demonstrate left and right rotation; four 90-degree rotations restore the previous orientation.

3. Demonstrate both up and down flips. Two vertical flips restore the previous orientation. Combine a rotation and a flip to show that actions use the current image.

4. Hold a direction to demonstrate repetition, then release it and confirm the image stops changing. Press center and confirm the application ends with the LEDs off.

5. Show that the application can be built and launched on the assigned Pi using the classroom deployment method. Retain the successful build and test evidence.

6 Submission checklist

Requirements specification - 25%: Submit this specification as a PDF, with any agreed changes to the controls or deployment method.

Demonstration - 25%: Demonstrate all user-accessible features above. The demonstration is evaluated on clarity (50%) and adherence to the specification (50%).

Source code and input data - 50%: Submit the documented main .m script, all three helper .m files, and the PNG used in the demonstration. Keep the files together so the application can be reproduced.